

KATE PORTALATIN

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Profile Summary

- **Senior Engineer** with experience working on **B2B Enterprise level SaaS** applications in fast paced startup like environments with a strong ability to demonstrate ownership and leadership as well as collaborate meaningfully in the **SDLC** (Software Development Lifecycle) by participating in the **agile** process.
- Designed and implemented **microservices** with **Kafka** and **Java** using **Spring** while architecting scalable, event-driven systems to support real-time data processing. Utilized **Kafka** for message brokering, enabling high-throughput event streaming and decoupled communication between services, ensuring system reliability and performance.
- Engineered dynamic, high-performance user interfaces using **React & Angular**, leveraging modern features like **React Hooks**, **Context API**, and **React Redux** for robust state management along with **NGRX** and **RXJS**. Focused on optimizing component-based architecture to ensure scalable, maintainable, and reusable code. Implemented advanced React patterns and best practices to improve code efficiency, enhance UI responsiveness, and streamline front-end development workflows, ensuring seamless integration with backend services and real-time data handling.

Education

Saint Joseph's University

M.S. Computer Science

Philadelphia, PA, 2012 - 2014

Art Institute of Pittsburgh

B.S. in Graphic Design

Pittsburgh, PA, 2006 - 2010

Technical Skills

Languages: Java, Python3, JavaScript, TypeScript, C, C++, COBOL, Assembly

Front End: React (including React Redux and React TypeScript), Angular, Next.js, Tailwind CSS, Redux, Context API, React Router, RxJS, React Testing Library, Styled-components, Sass, Scss

Back End: Java Spring Boot (Spring MVC, Spring Data JPA, Spring Security, Spring Cloud), Django RESTful APIs, Kafka, JUnit, Maven, Gradle, Node.JS, Express

Testing: Jest, Karma, JUnit, React testing framework, Cypress

Data: Azure Data Lake, Azure Data Warehouse, Snowflake, Jupyter notebook, Python, NumPy Pandas, Dataframes

Databases: MySQL, PostgreSQL, MongoDB, Elastic, Neo4j, Redis

Monitoring & Logging: ELK Stack (Elasticsearch, Logstash, Kibana), Prometheus, Grafana

Authentication & Authorization: JWT, OAuth 2.0, Okta

Dev Tools: Docker, Kubernetes, Git, CI/CD pipelines, Terraform, version management systems, configuration tools (Yaml, Ansible)

Servers: Centos, Debian, Ubuntu, Amazon Linux, running Java/Spring Boot, IBM mainframe systems (z/OS), and containerized microservices architectures

Cloud: Azure (primary expertise), AWS (EC2, Lambda, S3, RDS, SQS, CloudFormation), GCP (Compute Engine, Kubernetes Engine, Cloud Functions)

Professional Experience

Blue Yonder *Senior Software Engineer*

Coppell, TX (Hybrid) * Mar 2020 – Dec 2024

- Architected and maintained a dynamic front-end ecosystem using **React**, **Next.js**, and **Node.js**, leveraging **Redux** for centralized state management through **Actions**, **Stores**, and **Reducers** to ensure efficient data flow and consistent UI updates. Developed **reusable** React **components** with **TSX**, employing **React Hooks** (**useEffect**, **useState**, **useMemo**) to manage side effects and optimize performance. Integrated secure backend services with seamless communication, ensuring robust performance, real-time data updates, and near-perfect uptime across the platform. Utilized modern React patterns and best practices to build scalable and maintainable front-end solutions that enhanced user experience.
- **Mentored** new hires during onboarding by providing hands-on training in core technologies and development workflows, accelerating their integration into the team and enabling them to contribute effectively to key projects.
- **Managed infrastructure as Code** using **Terraform** for **configuration and resource management**.
- **Spearheaded** a cross-functional initiative to align multiple departments, **coordinating** closely with **Product**, **DevOps**, **IT**, and other teams to optimize infrastructure environments, resulting in a significant increase in developer productivity and **maintainability**. Advocated strongly for best practices, including CI/CD pipelines, automated testing, and code reviews, to ensure consistent code quality, faster delivery, and reduced manual intervention across the organization.
- Employed **ElasticSearch**, **Grafana**, and **Prometheus** to continuously monitor **ingestion** rates, **transaction** volumes per second, and system health, proactively tracking performance metrics to ensure system stability and prevent **downtime**. Integrated the **ELK** stack (**Elasticsearch**, **Logstash**, and **Kibana**) for efficient log aggregation, analysis, and visualization, enabling real-time monitoring and insights into system performance and errors.
- Worked with and gained knowledge of **Azure IAM** for role-based access control (**RBAC**) and managing secure data storage using **Azure Key Vault** to protect sensitive environment variables.
- Designed and managed Azure **Application Gateway** for efficient Layer 7 load balancing and secure traffic routing, utilizing the **Round Robin** algorithm for even distribution of traffic across instances and **Consistent Hashing** to maintain session persistence. Leveraged **Azure Functions** for **serverless** computing and **workflow automation**, enabling scalable, **event-driven** architectures and reducing infrastructure overhead.
- Designed and implemented **event-driven** microservices architecture using **Apache Kafka** as a **message broker** for **asynchronous** messaging, enabling real-time data processing and scalable, decoupled system interactions. **Kafka** was leveraged for reliable **event streaming**, high-throughput data flow, and managing **Kafka topics** for event communication between **microservices**. **Consumer groups** were utilized to ensure parallel processing and message ordering. The system architecture supported event-driven workflows, allowing each service to independently scale while maintaining eventual consistency.
- Orchestrated and managed **Kubernetes clusters** for **containerized applications**, utilizing **Docker** for containerization and ensuring seamless deployment across environments. Employed **Kubernetes** features like **Pods**, **Deployments**, **ReplicaSets**, and **Namespaces** for efficient scaling, high availability, and resource management. Leveraged **Helm** to automate the deployment and management of **Helm charts**, simplifying application versioning, rollback, and updates while ensuring consistency in multi-environment setups. Utilized ConfigMaps, Secrets, and Persistent Volumes for managing configuration, sensitive data, and stateful applications effectively.
- Designed and developed **microservices** in **Java** using **Spring Boot**, leveraging **Spring Data JPA** for efficient database management and **Spring Security** to ensure secure communication between services. Utilized **Spring Cloud** for building scalable, cloud-native microservices and **Spring MVC** for creating RESTful APIs.

Applied **Dependency Injection** for better modularization and maintainability and employed **JUnit** and **Spring testing** to ensure robust unit and integration testing. Worked with **Java 8** features, including **streams** and **lambdas**, to enhance code performance and readability. Collaborated in building **front-end components with React** and managing system configurations to ensure seamless integration and efficient communication across services.

- Consistently applied **SOLID** principles in software design to build modular, scalable, and maintainable systems. This included practices like **single responsibility** for modular design, **open/closed** architecture to allow easy extension while minimizing modifications, **Liskov substitution** to ensure derived classes could be used interchangeably with base classes, and **dependency injection** to decouple components and improve testability, all of which contributed to reducing technical debt and fostering long-term codebase stability.
- Led initiatives by effectively obtaining buy-in from **stakeholders**, including **product owners**, **project managers**, and **executives**, ensuring alignment with business goals and securing necessary resources for successful project execution and delivery.

Core Logic *Software Engineer*

Coppell, TX * May 2019 -Feb 2020

- Integrated advanced React features by leveraging **Redux** for robust state management and **React Query** for reactive data streams, ensuring dynamic, real-time UI updates and enhanced user responsiveness. Developed highly reusable and modular **React components**, utilizing **React Hooks** and **useEffect** for managing side effects, streamlining front-end development, and improving overall application performance.
- Integrated **JWT** for secure, token-based authentication in **React** applications, enabling role-based access control (**RBAC**) and ensuring secure, stateless communication between the client and backend services. Worked extensively with **React Router** to create seamless authentication flows, ensuring that user access is restricted based on roles and maintaining a high level of security across the platform.
- Championed cross-functional collaboration among **development**, **QA**, and **operations** teams, aligning efforts with strategic business goals while leveraging cloud platforms like **Azure** and **AWS**. Specifically, utilized **AWS S3** buckets for secure, scalable file storage, ensuring efficient management and retrieval of assets within the microservices environment. Integrated these cloud services into both **React** front-end applications and **Java Spring Boot** backend services to ensure smooth interactions and scalability.
- Applied object-oriented programming (**OOP**) principles in **Java Spring Boot**, including **encapsulation**, **inheritance**, and **polymorphism**, to build modular, reusable, and maintainable backend services that drive robust solutions. Developed RESTful APIs that interact seamlessly with the front-end, ensuring smooth data flow between the **React / Angular** UIs and the backend services.
- Worked with **Angular** on certain projects for legacy system modernization, designing dynamic, data-driven interfaces and integrating **RxJS** for handling asynchronous data streams, ensuring that the user interfaces remained responsive and intuitive while seamlessly interacting with **Java Spring Boot** backend services.

Choptank Transport *Software Engineer*

Preston, MD * Aug 2016 – Mar 2019

- Participated in the greenfield implementation of logistics solutions using **Node.js** with **Express** and **RESTful APIs**, reducing manual processes and improving software scalability to handle higher transaction volumes with minimal latency.
- Developed backend services using **Spring Boot**, leveraging its robust ecosystem for building scalable and maintainable microservices, ensuring smooth integration with the **Node.js** environment.

- Designed and implemented **Angular-based** dynamic web interfaces, ensuring real-time, responsive UIs for users while integrating seamlessly with backend services, including **Spring Boot** APIs, to provide a consistent, smooth user experience across both the **frontend** and **backend**.
- Collaborated with cross-functional teams to define requirements and ensure seamless interaction between the **Node.js** backend, **PostgreSQL** data layer, and **Angular** front-end, resulting in an intuitive and efficient logistics management platform.

Highmark Blue Cross Blue Shield *Application Developer*

Wilmington, DE * Jan 2015 – Aug 2016

- Integrated AngularJS with legacy healthcare applications, developing modern web interfaces to enhance user experiences while ensuring data integrity and system stability.
- Utilized Java to support backend services, ensuring seamless data synchronization and improved functionality for healthcare professionals.